Mechanisms underlying movement patterns of the South American tortoise Chelonoidis chilensis

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Abstract

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Introduction

Animal trajectories are the result of a complex interplay between internal and external factors, including the animal's physiological state, environmental conditions, and interactions with other individuals. Understanding the mechanisms underlying movement patterns is crucial for predicting animal behavior, managing wildlife populations, and conserving biodiversity. How animals take decisions about when and where to move is a fundamental question in ecology and behavioral biology. Movement patterns can be influenced by a variety of factors, including resource availability, predation risk, social interactions, and environmental conditions. In this study, we focus on the South American tortoise Chelonoidis chilensis, a species that is classified as endangered. As a reptile, it is ectothermic and relies on external sources of heat to regulate its body temperature. This makes it particularly sensitive to environmental conditions, which can influence its movement patterns and behavior. By performing comuter simulations is possible to explore the effects of different factors on movement patterns and to identify candidate mechanisms that drive these patterns. Previous studies showed that tortoises movement is subdiffusive which suggests....

By analyzing GPS tracking data collected over several camàigns, we aim to uncover the statistical properties of their movement patterns and identify potential factors influencing their behavior.

Methodology

The study was carried out in a site of approximately 25 hectares, located about 20 km north of the city of San Antonio Oeste, Province of Río Negro, Argentina, where there is a stable population of C. chilensis with a large number of specimens. The area in general belongs to the Monte Austral phytogeographic unit, characterized by a shrub steppe with a predominance of Larrea spp. with several strata, the lower one being grasses and herbs, all with very little coverage, and particularly with very few cacti (Oyarzabal et al. 2018). The specific study area is characterized by a vegetation cover, mostly with xerophytic characteristics, with a predominance of grass clumps (Poaceae) and shrubs (León et al. 1998, Morello et al. 2012). In turn, the grass stratum has a higher species richness than the other Zygophyllaceae of the Monte steppes. The soil is rather sandy with a low percentage of clay and abundant pebbles. Close to the border of the study area, there is the Gran Bajo del Gualicho, a Patagonian plain of up to 70 meters below sea level, characterized by a high saline concentration. This is why there are more saline sectors, where we can find halophyte species such as Suaeda divaricata and several species of the genus Atriplex spp. Scarce precipitations give an average of approximately 255 mm in the year, with minor peaks in spring and fall. Annual average temperature is around 14.5 °C with significant daily thermal amplitudes and a persistence of the west and southwest winds, making the region more arid (Godagnone and Bran, 2008). The minimum average temperature is 1°C, and the maximum is 30 °C, with extreme recorded temperatures of -11.5 and 44.6 °C (data for 1961–2021 provided by the National Meteorological Service of Argentina). We tracked six tortoises continuously during a whole year using GPS data loggers (i-gotU GT-120 Mobile Action, Taiwan), recording geographical coordinates every 15 minutes. The positions were recorded continuously from 6am to 10pm and from 2am to 3am. The equipment was attached to the carapace of each individual using a camouflage tape (@Duck Tape, Real Tree Hardwood Camouflage) to mimic the environment, with the precaution of not leaving any sharp corners, in order to prevent the tortoise from getting trapped in the vegetation. After attaching the device, tortoises were released at the same place that were found. The weight of the equipment never exceeded

10In this work we present analysis and results spanning a whole year, involving a total of 20149 hours of GPS data loggers recording. Those recorded geographical positions corresponded to 813 trajectories from 6 individuals. The GPS data loggers were charged every fifteen days except during the brumation period, when animals were not disturbed and were observed every two weeks to check if they remained brumating at the same location. All recorded trajectories were preprocessed to exclude those corresponding to speeds above a threshold of 15 m/min, assessed as a reasonable maximum speed for the species, according to the speed empirical distribution presented in Fig.S3. The distance traveled as a function of time was calculated from the (latitude, longitude) data using the Haversine function. For each behavioral period, we calculated the mean square displacement (MSD) as a function of time, defined as explained in the supplement and was averaged over all daily mesasurements. Additionally, accelerometer signal was classified into two states: walking and resting, using the random forest classification algorithm in Python and we calculated the distribution of step lengths and waiting times for each state. The distributions were fitted using maximum likelihood estimation, for each individual and behavioral period. The results were analyzed to identify patterns and differences between individuals, sexes, and behavioral periods. Here explain that accelerometer has signal every 1,7s while GPS has signal every 15 min. So we aggregated acc data to 15min following the crteria.....

Results

The distances traveled by the tortoises were measured every 15 minutes. The probability distribution of traveled distances, $P(d)$, for all tortoises is shown in Figure 4. It exhibits a power-law behavior over a wide range of distances, with an exponent $\beta \approx 2.1(2)$. This indicates that the tortoises' movement patterns are characterized by a heavy-tailed distribution, suggesting that they occasionally make long-distance movements interspersed with shorter ones.

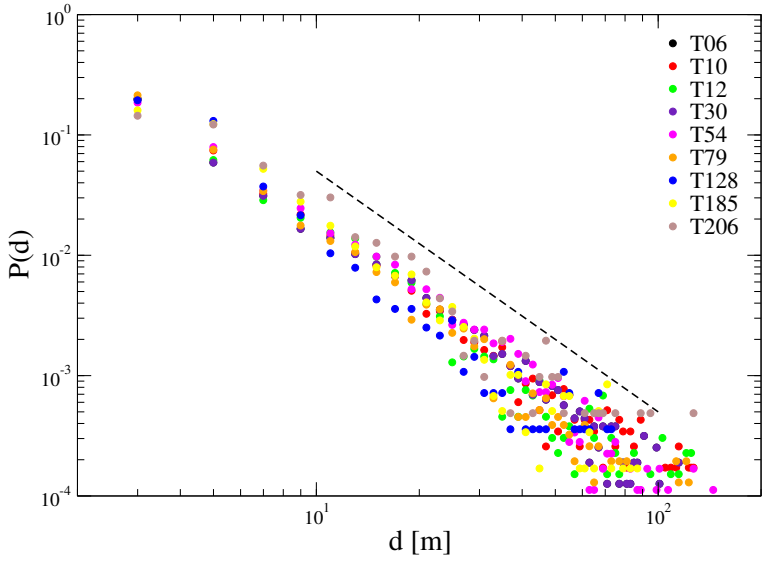


Figure 1: Distribution of traveled distances every $\Delta t = 15min$ during the day (9:00 a.m. to 5:00 p.m.).

The mean square displacement (MSD) as a function of time for all tortoises at a daily scale, calculated from the recorded trajectories, can be seen in Figure 2. As tortoises behavior changes according to the diferent periods: November–January (reproduction), February–March (laying eggs), and April–October (brumation), we computed the MSD separately for each behavioral period. The MSD exhibits a power-law behavior, with an exponent α that varies between subdiffusive ($\alpha < 1$) and diffusive ($\alpha \approx 1$), however differences in the slope of the power law are not markedly different (see Figure **??**). This suggests that the tortoises' movement patterns are not purely random but are influenced by underlying behavioral mechanisms.

According to Figure 2, the MSD exhibits a power-law behavior during the February–March period, with an exponent $\alpha \approx 1.2(1)$, indicating superdiffusive behavior. In contrast, during the November–January and April–October periods, the MSD shows subdiffusive behavior, with exponents $\alpha \approx 0.8(1)$ and $\alpha \approx 0.9(1)$, respectively. These results suggest that the tortoises' movement patterns vary across different behavioral periods, potentially reflecting changes in their activity levels and environmental conditions.

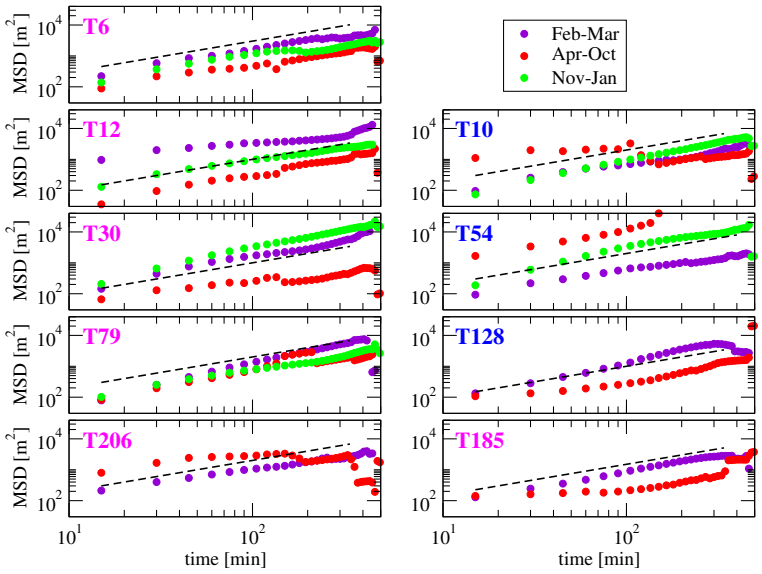


Figure 2: MSD versus time for all tortoises (labeled as T_{number}) from GPS recordings during daylight hours (9:00 a.m. to 5:00 p.m.), categorized by behavioral period as indicated. The color of each label indicates the turtle's sex (magenta for females and blue for males). Dotted black lines represent a slope corresponding to diffusive behavior ($\alpha \approx t^1$).

According to Table S4 the MSD exponent α is consistently less than 1 for all tortoises, indicating subdiffusive behavior. However some individuals exhibit exponents significaly lower that one wich is indicative of a more constrained movement pattern. The distribution of traveled distances, $P(d)$, also shows variability among individuals, with some exhibiting heavier tails than others. This suggests that while the overall movement patterns of the tortoises are subdiffusive, there are individual differences in their movement strategies and behaviors.

Figure 2 shows the MSD behavior separated by males and females. The exponent α (Eq. S1) does not appear to change significantly between males and females.

A similar situation is observed for the distribution of traveled distances (Figure **??**), although in this case the dispersion in the curves may be larger among females than among males.

The distribution of step lengths and waiting times from accelerometer data is shown in Figures 4 and 4. The step length distribution exhibits a heavy-tailed behavior, indicating that the tortoises occasionally make long steps interspersed with shorter ones. The waiting time distribution also shows a heavy-tailed behavior, suggesting that the tortoises may have periods of inactivity interspersed with bursts of activity. These distributions provide insights into the tortoises' movement patterns and can inform the development of models to describe their behavior.

Modeling Tortoise Movement Patterns

Inspired in those measurements we proposed a model based on a *Stop and Go* random walker moving in a homogeneous continuous space. We simulate $N \approx 100.000$ individuals with no interaction between them, each alternating between two states: movement (*Go*) and rest (*Stop*).

We assume that each tortoise performs random walks with constant speed interspersed with resting periods, provided the duration of the walking bouts is distributed with a power-law, or Pareto, distribution,

$$P_b(t) = \frac{\alpha - 1}{t^\alpha}, \qquad t \geq 1,$$

where $P_b(t)\,dt$ is the probability that a bout of walking will last from time $t = 0$ to a time between t and $t + dt$.

At each step, we draw the total time to be spent in a state directly upon entering it from its full duration distribution. To this end, we use inverse transform sampling (*inverse-CDF sampling*): given a probability density $f(t) \propto t^{-\alpha}$ truncated between a minimum time t_{min} and a maximum time t_{max}, its cumulative distribution function can be inverted to obtain a closed-form expression that, given a random number u uniformly distributed in $[0, 1]$, directly yields the sampled time:

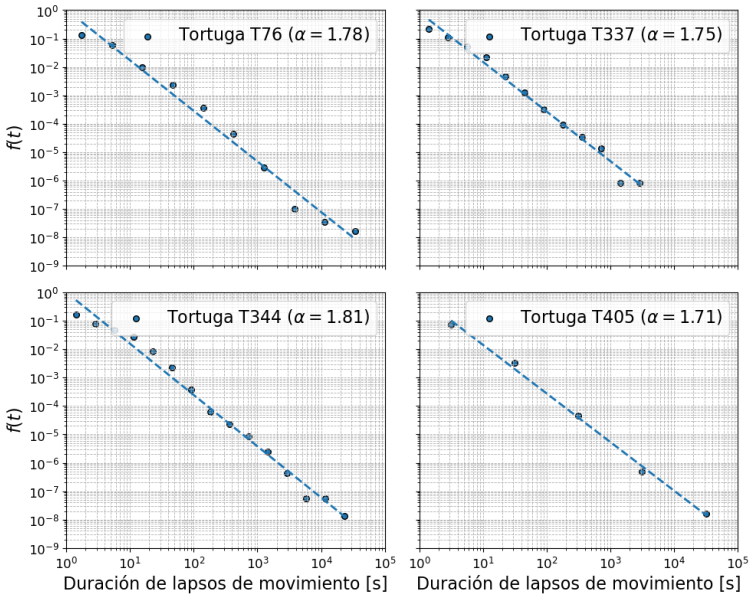


Figure 3: Distibution of steplenghts from accelerometer data.

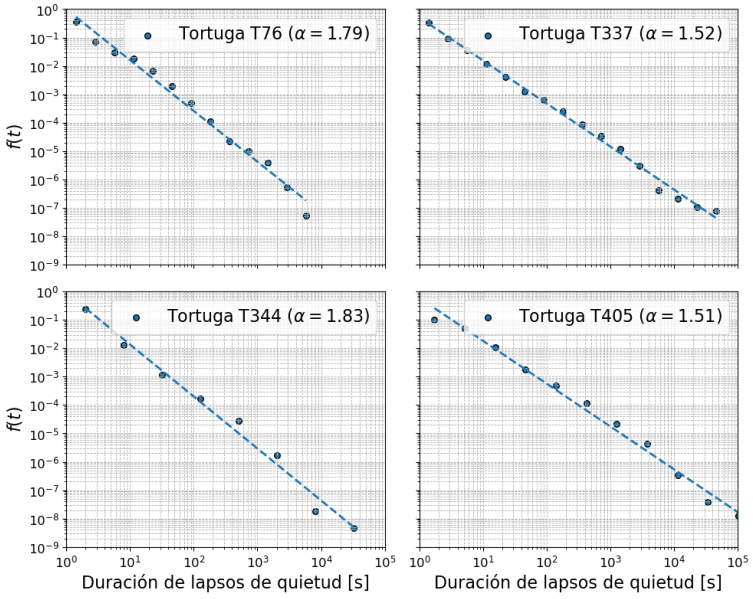


Figure 4: Distibution of waiting times from accelerometer data.

$$t = \left[(t_{max}^{1-\alpha} - t_{min}^{1-\alpha})\, u + t_{min}^{1-\alpha} \right]^{\frac{1}{1-\alpha}} \tag{1}$$

This method yields, in a single operation, the full time that an individual will spend in the movement or rest state, without having to evaluate the transition probability step by step. Since we simulate a large number of individuals over long time spans, we reduce computational cost. The individual moves at discrete time intervals δt, taking steps of constant length L. The direction of each step follows a correlated random walk: the new orientation angle θ_{i+1} is obtained from the angle of the previous step, θ_i, by adding Gaussian noise with zero mean and standard deviation σ_θ,

$$\theta_{i+1} = \theta_i + \eta_i, \qquad \eta_i \sim \mathcal{N}(0, \sigma_\theta^2). \tag{2}$$

The position of the individual is therefore updated at each time step according to

$$x_{i+1} = x_i + L\cos(\theta_{i+1}), \qquad y_{i+1} = y_i + L\sin(\theta_{i+1}). \tag{3}$$

This angular correlation introduces directional persistence into the trajectory: unlike a simple random walk, in which the direction is drawn independently at each step, the individual tends to maintain its heading over short time scales, producing locally straight trajectories that curve gradually as angular perturbations accumulate.

In the *Stop* state, by contrast, the position of the individual remains fixed, $x_{i+1} = x_i$ and $y_{i+1} = y_i$, and the only process that advances is the count of the time elapsed in that state.

The duration of each bout, both *Go* and *Stop*, is drawn upon entering the corresponding state using the inverse transform sampling method described above, from the truncated power-law distribution associated with each of the two states.

Computer simulations

We performed simulations of the model described above, exploring three different configurations. In the first configuration, we kept the walking time constant and let the rest time follow a power law but we couldn't reproduce the observed behavior in the step length distribution nor in the MSD.

Finally we propose a model in which both time distributions follow a power law, $f_S(t) \propto t^{-\alpha_S}$ for rest and $f_W(t) \propto t^{-\alpha_W}$ for walking, with $\alpha_S, \alpha_W > 0$. To explore the parameter space, we performed a sweep over $\alpha_S \in [0.5, 2.0]$, $\alpha_W \in [1.0, 3.0]$ and $\sigma_\theta \in [0.2, 2.0]$, computing in each case the 15-minute displacement distribution, $P(d) \propto d^{-\alpha_d}$. As an example, Figure S13 shows these distributions for different values of α_S, with $\alpha_W = 1.75$ and $\sigma_\theta = 0.2$ held fixed. Across the whole sweep, the decreasing power-law regime ($\alpha_d > 0$) appears only when the rest distribution decays more slowly than the walking distribution, that is, when $\alpha_S < \alpha_W$.

Within the region $\alpha_S < \alpha_W$, we searched for combinations that reproduce the empirically observed behavior. Figure 5 shows the best configuration obtained, with $\sigma_\theta = 0.25$, $\alpha_W = 1.8$ and $\alpha_S = 1.1$, that reproduces the empirical exponent, $\alpha_d \approx 2$, between $d \in [20, 200]$ (u.a.).

As in the previous case, once the displacement distribution had been reproduced, we analyzed the MSD of this model, shown in Figure 6. The MSD displays ballistic behaviour at short time scales, with an exponent that transitions to $\beta = 1$ at long time scales.

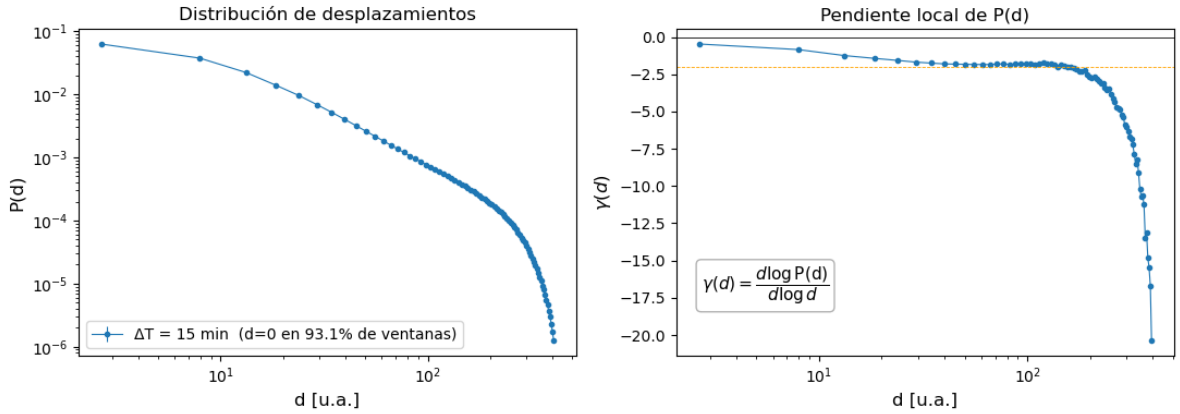


Figure 5: 15-minute displacement distribution for $\sigma_\theta = 0.25$, $\alpha_S = 1.1$ and $\alpha_W = 1.8$.

In the Figure 7, we show three examples of trajectories with the best exponents with this configuration.

Statistical interpretation

Another approach consists of defining an instantaneous transition rate $\gamma(t)$, a function of the time already spent in the current state, and evaluating it at every discrete time step δt of the simulation to decide whether the individual switches state. Although this approach naturally accommodates a time-dependent rate, it requires evaluating a transition probability at every time step of each bout, which considerably increases the computational cost when simulating a large number of individuals.

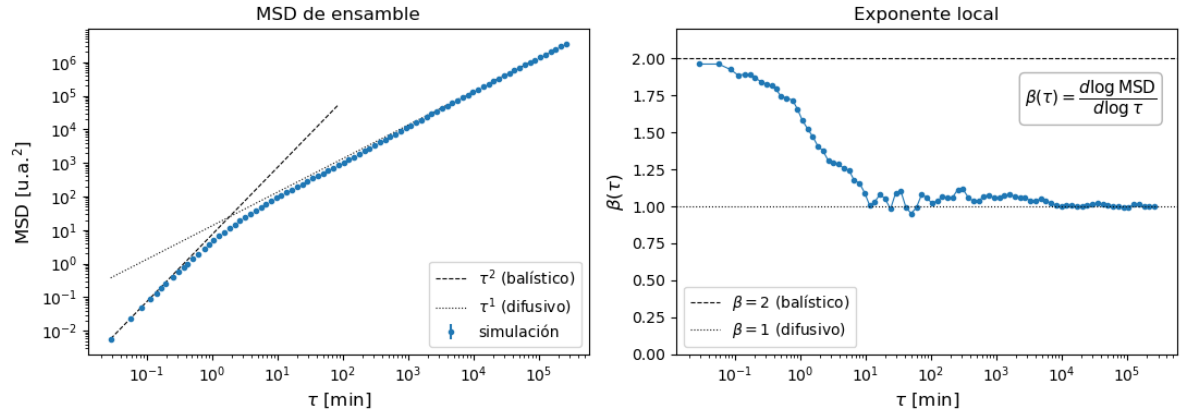


Figure 6: MSD for $\sigma_\theta = 0.25$, $\alpha_S = 1.1$ and $\alpha_W = 1.8$. The gray lines indicate balistic and diffusive behavior.

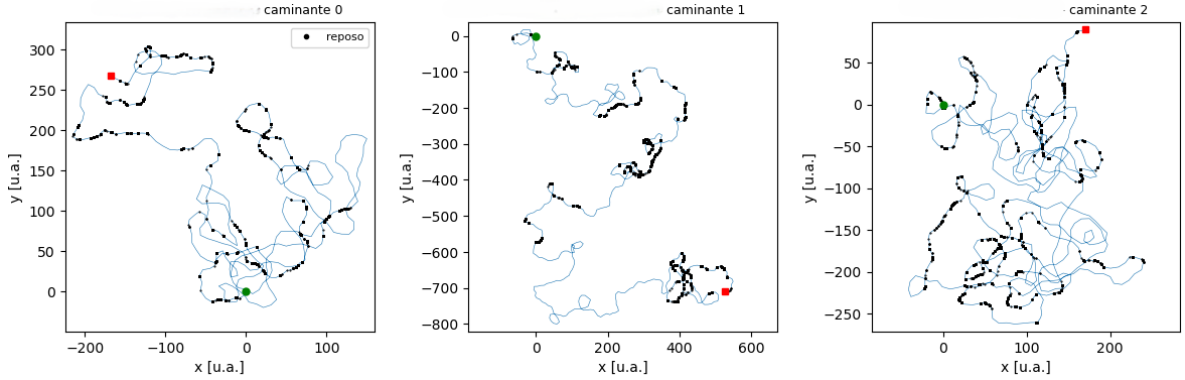


Figure 7: Three examples of trajectories for $\sigma_\theta = 0.25$, $\alpha_S = 1.1$ and $\alpha_W = 1.8$.

This behaviour can also be explained by considering the transition probability from walking to resting, $\gamma(t)$. The transition probability is the probability that the tortoise will stop walking at a time between t and $t + dt$, given that it is walking at time t, i.e.

$$\gamma(t) = \frac{P_b(t)}{\text{P. of walking at time } t} = \frac{P_b(t)}{1 - \text{P. of having stopped between 0 and } t} = \frac{P_b(t)}{1 - \int_0^t P_b(t')\,dt}.$$

Since $\int_1^t (\alpha - 1)t^{-\alpha} = 1 - t^{1-\alpha}$, we obtain

$$\gamma(t) = \frac{\alpha - 1}{t}, t \geq 1,$$

i.e.\ the probability that the tortoise will decide to rest gets smaller the longer it has been walking.

Power-law-distributed bouts of behaviour have been discussed before [**?**], [**?**], but not everyone seems to agree on power laws [**?**].

Transition rate on a fixed maximum behaviour time

It has been shown that tortoises if can be clasified into two groups, in which both follows a power law distributions, $f(t)$, of exponent α with a maximum time of behaviour, t_{max}, [**?**]. In order to facilitate the theoretical analysis, we can infer a transition rate from one state to another, which is a function of the time spent in the current state, t. Another thing to consider is that in our simulation we have discrete time steps, δt. We will obtain the transition rate, through the following relation:

$$\gamma(t) = 1 - P(f(t + \delta t)|f(t)) \tag{4}$$

Where $P(f(t + \delta t)|f(t))$ is the conditional probability of being in the same state at time $t + \delta t$ given that it was in that state at time t. This conditional probability can be expressed as:

$$P(f(t+\delta t)|f(t)) = P(f(t)|f(t+\delta t))\frac{P(f(t+\delta t))}{P(f(t))} \tag{5}$$

in which $P(f(t)|f(t+\delta t)) = 1$ and $f(t) = At^{-\alpha}$ with A a normalization constant. $P(f(t))$ can be calculates as follows:

$$P(f(t)) = \int_t^{t_{\max}} f(t')\,dt' = \frac{A * \left(t_{\max}^{1-\alpha} - t^{1-\alpha}\right)}{1-\alpha} \tag{6}$$

in this equation we are assuming that the state began at time $t = 0$ so if we apply this to the conditional probability we have:

$$P(f(t+\delta t)|f(t)) = \frac{t_{\max}^{1-\alpha} - (t+\delta t)^{1-\alpha}}{t_{\max}^{1-\alpha} - t^{1-\alpha}} \tag{7}$$

Finally, we can express the transition rate as:

$$\gamma(t) = \frac{(t+\delta t)^{1-\alpha} - t^{1-\alpha}}{t_{\max}^{1-\alpha} - t^{1-\alpha}} \tag{8}$$

Discussion and Conclusions

The seasonal periods still need to be refined. Here, only the calendar months were considered, but factors such as temperature, late spring conditions, and related environmental variables should probably also be taken into account. This convergence to a difussive behaviour is mainly due to the maximum time for the power law distribution, which lets the difussive behaviour of the gaussian distribution of the angular variation to control the MSD functionality at longer times. Also the maximum distribution times allows for several switches between the states, if we had no maximum times, due to the exponents of the distribution we are analysing, we would have little movement of the tortoises.

1 Acknowledgements

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Supplement

Para todas las tortugas el MSD Figure **??** y la distribucion de distancias $P(d)$ Figure S1

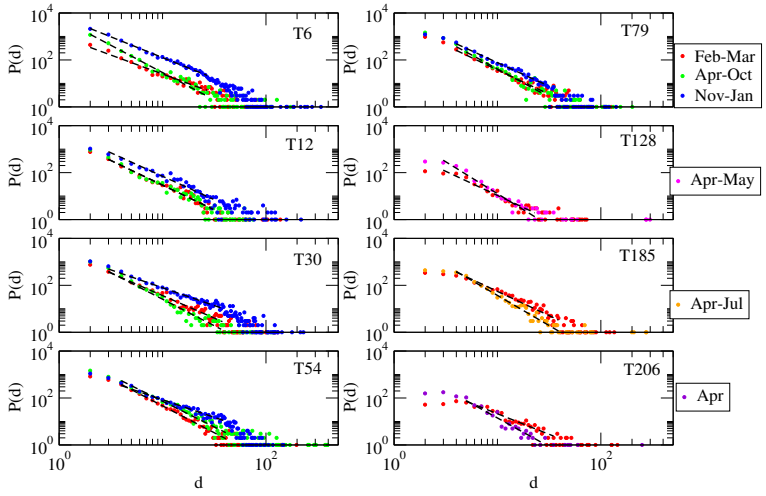


Figure S1: Distribucion de distancias recorridas en intervalos de 15min segun separacion estacional para todas las tortugas

In both figures, Fig. **??** and Fig. S1, the dotted lines correspond to the best fits obtained. Table 1 lists the values of these exponents ($MSD(t) \approx t^{\alpha}$ and $P(d) \approx d^{-\beta}$). As can be seen, there is considerable dispersion among turtles and across different periods of the year.

It should be noted that for some tortoises (T128, T185, and T206) only limited data are available. For example, no data are available for the November–January period, and for the April–October period only data from the first months (April and May) are available, as indicated in Figures **??** and S1.

Figure **??** shows all the measured quantities for turtle T10. There are some temporal gaps corresponding to missing data.

A similar behavior is observed for the other turtles (T6, T12, T30, T54, T79, T128, T185, and T206).

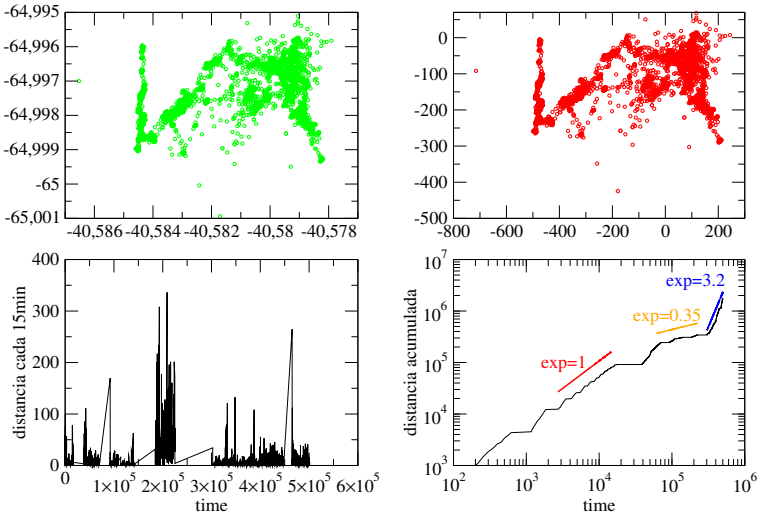


Figure S2: Datos de la Tortuga T10: A) Datos crudos de (lat,long) ; B) Lo mismo que en A) pero metros, para chequear que el pasaje de datos funcione. C) Distancia recorrida (cada 15min) en funcion del tiempo; D) Distancia acumulada vs time: los colores indican los exponentes del ajuste en cada periodo.

By testing a different temporal interval, $\Delta t = 30\,\text{min}$, the exponent does not change, nor is any improvement observed in $P(d)$ (Figure S6).

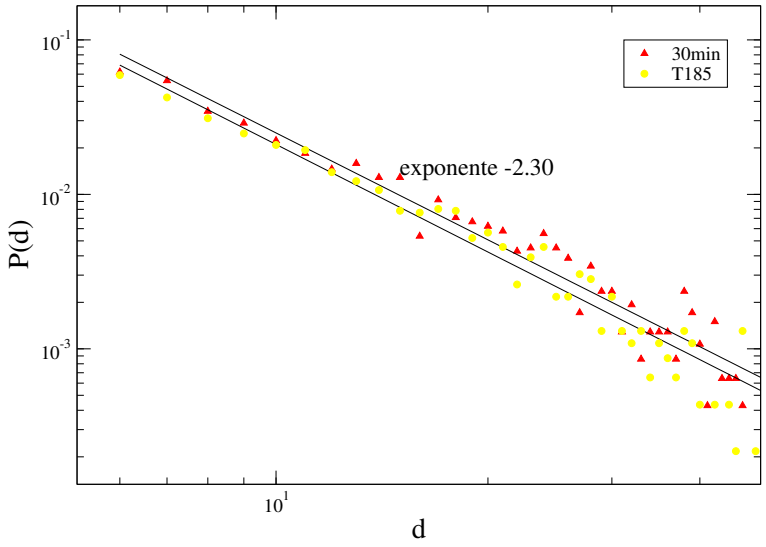


Figure S3: Distribution of traveled distances for T185, using $\Delta t = 15\,\text{min}$ and $\Delta t = 30\,\text{min}$.

Exponente alfa (MSD)				Exponente beta (P(d))			
Tortuga	**Feb-Mar**	**Abril-Oct**	**Nov-Ene**	**Tortuga**	**Feb-Mar**	**Abril-Oct**	**Nov-Ene**
T6	0.269	0.915	0.565	**T6**	1.62	2.3	1.76
T10	0.064	0.539	0.883	**T10**	1.8	2	2.4
T12	0.34	0.67	0.262	**T12**	2.1	2.1	1.99
T30	0.774	-0.117	0.708	**T30**	1.92	2.2	1.51
T54	0.901	0.001	0.629	**T54**	2.29	2.2	1.5
T79	0.364	0.075	0.292	**T79**	2.08	2.4	2.01
T128	0.588	0.865	S/D	**T128**	2.1	2.7	
T185	0.651	0.367	S/D	**T185**	2.1	2.5	
T206	0.792	0.54	S/D	**T206**	1.75	2.4	
Valor medio	0.5(3)	0.4(4)	0.5(2)	**Valor medio**	2.0(2)	2.3(2)	1.9(3)

Figure S4: Table 1: Exponents of MSD and $P(d)$ for each turtle and for each period. Males and females are distinguished by color, and the exponents were calculated using all available data.

Mean square deviation (MSD)

We can fit an exponent:

$$\text{MSD} \approx t^{\alpha} \tag{S1}$$

Based on the observed behavior, we can define an exponent as:

$$\text{P(d)} \approx d^{-\beta} \tag{S2}$$

The mean exponent for the 9 turtles is: $\beta \approx 2.1(2)$, as shown in Figure **??**.

By testing a different temporal interval, $\Delta t = 30\,\text{min}$, the exponent does not change, nor is any improvement observed in $P(d)$ (Figure S6).

S0.1 First configuration: constant walking time

Figure S10 shows several displacement distributions, with $\sigma_\theta = 0.2$ and a constant walking time $T_{\text{walk}} = 10$ t.s. (time steps, i.e., 10 simulation steps) for all three curves. What varies between the curves is the exponent of the rest-time distribution.

In this figure, we observe that the displacement distributions exhibit characteristic distances that are approximately multiples of $L\,T_{\text{walk}}$. Because the model adds Gaussian noise to the direction of motion, these characteristic values lie slightly below the aforementioned multiples, since the exact value is only attained when the individual moves in a straight line. These peaks correspond to a given number of walking bouts completed within the 15-minute window; for example, the first peak corresponds to a single walking bout, after which the individual remained at rest for the remainder of that window. As a consequence, these characteristic distances become blurred as the rest-time exponent α_S increases, since long rest times become less likely.

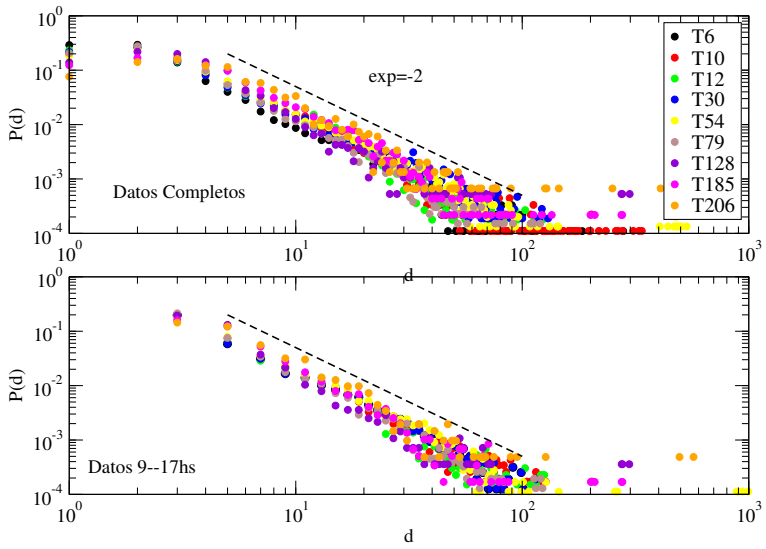


Figure S5: Distribucion de distancias recorridas cada $\Delta t = 15min$.

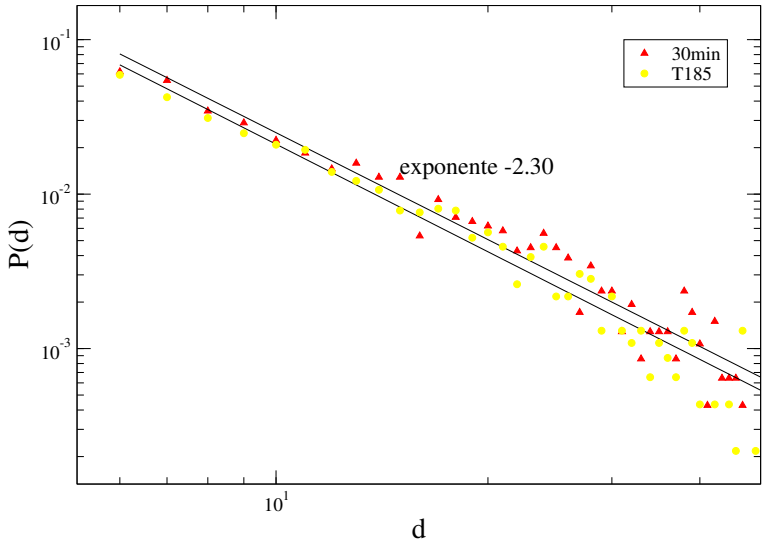


Figure S6: Distribution of traveled distances for T185, using $\Delta t = 15$ min and $\Delta t = 30$ min.

Given the discrepancy between the distributions obtained and the empirical distribution, we conclude that this model cannot reproduce the behavior observed in the tortoises, and we therefore discard it.

S0.2 Second configuration: constant walking time

In the second configuration, we kept the rest time constant and let the walking time follow a power law. The displacement distribution for different rest times is shown in Figure S11, keeping the walking-time exponent and σ_θ fixed.

Unlike the first model simulations, the distributions display power-law behavior, and they converge to the same distribution once the rest times exceed the duration of the analyzed window (15 minutes). This configuration reproduces the power-law behavior with exponent $\alpha_d \approx 2$ observed in the tortoises for rest times longer than 500 seconds. The MSD produced with this model is shown in Figure S12.

All the simulations exhibit a superdiffusive behavior at the begining and crosses over to diffusive behavior at longer times.

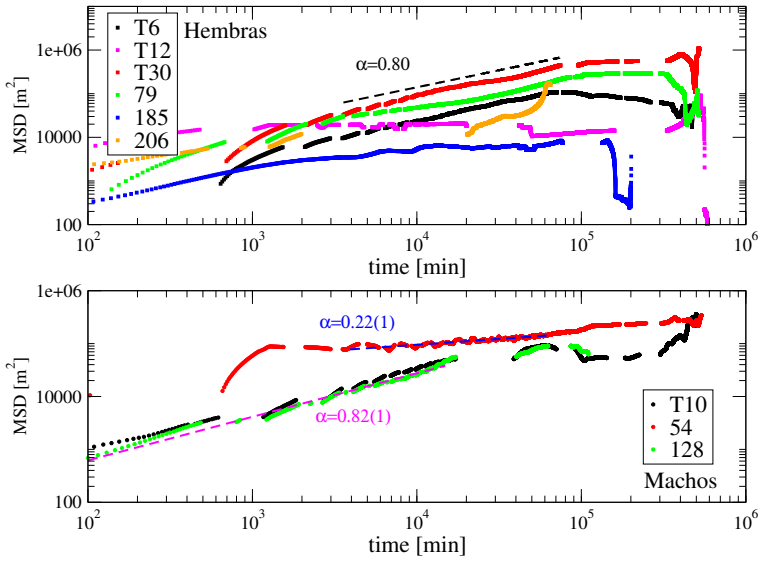


Figure S7: MSD separated in males and females

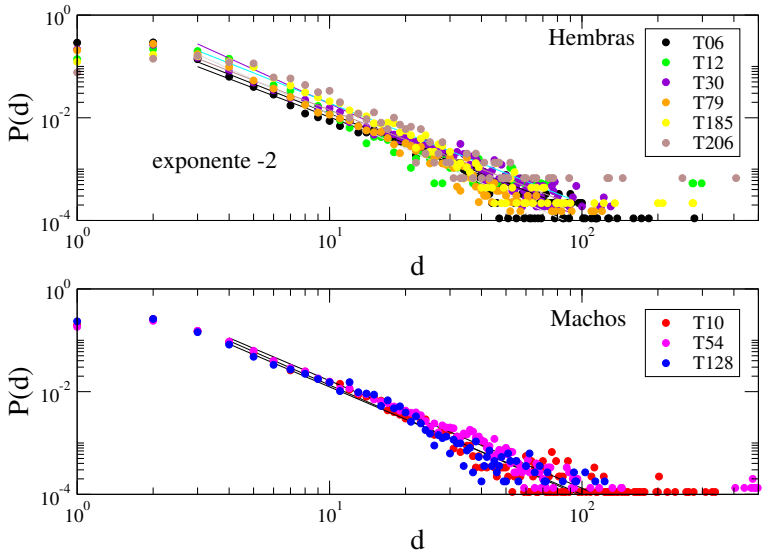


Figure S8: P(d) vs d separated in males and females

S0.3 Third configuration: constant walking time

		α (MSD)	β (P(d))
Hembras	T6	0.83	1.88
	T12		1.92
	T30	0.78	1.91
	T79	0.74	2.22
	T185	0.79	2.22
	T206	0.53	1.95
Machos	T10	0.82	2.16
	T54	0.22	2.09
	T128	0.82	2.09

Figure S9: Table 2: MSD and $P(d)$ exponents for each tortoise, separated by males and females, calculated using all available data without separation by periods.

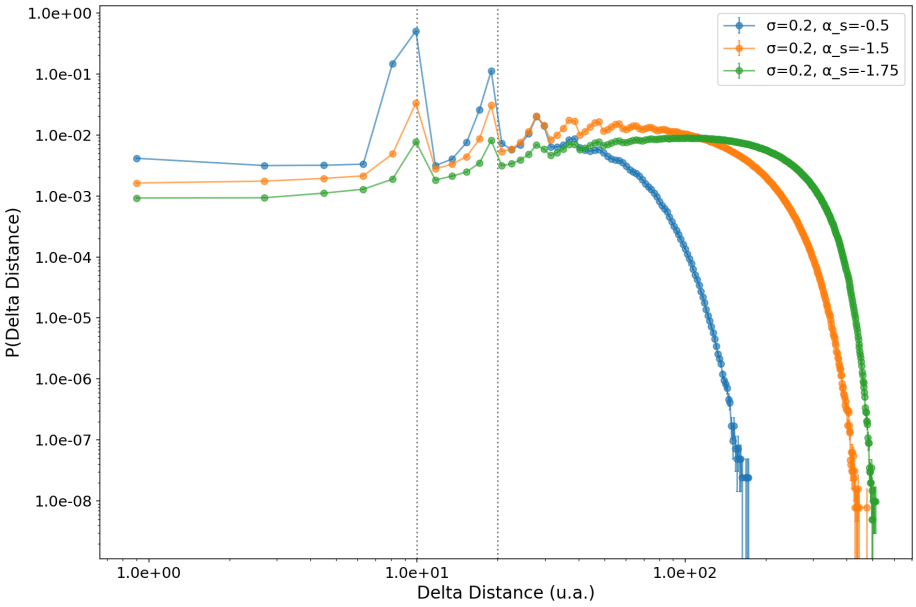


Figure S10: Displacement distribution for the first model, with constant walking time.

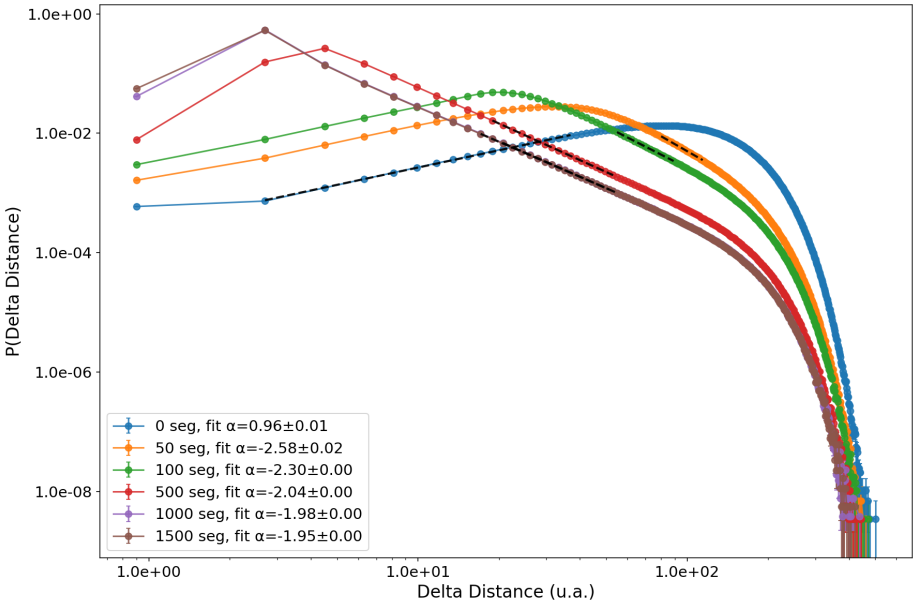


Figure S11: Displacement distribution for the second model, with constant rest time.

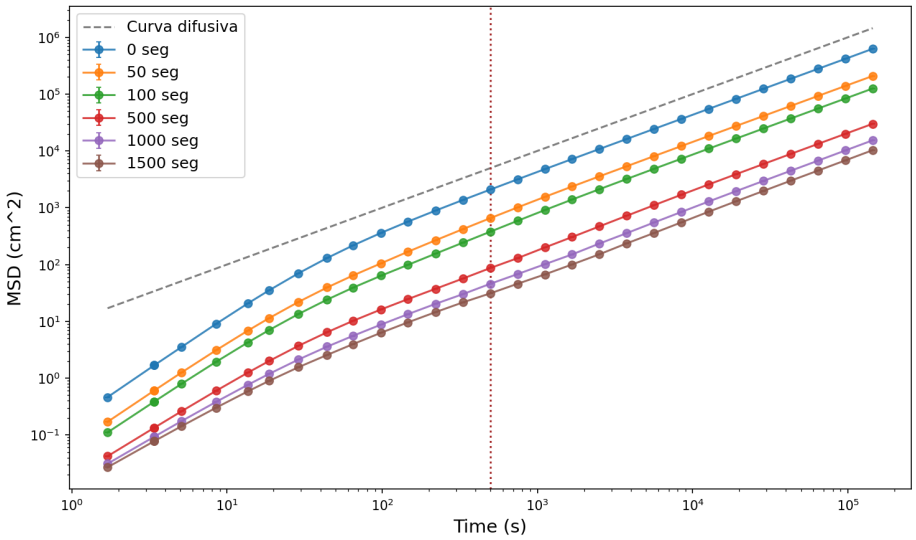


Figure S12: MSD for the second model, with constant rest time.

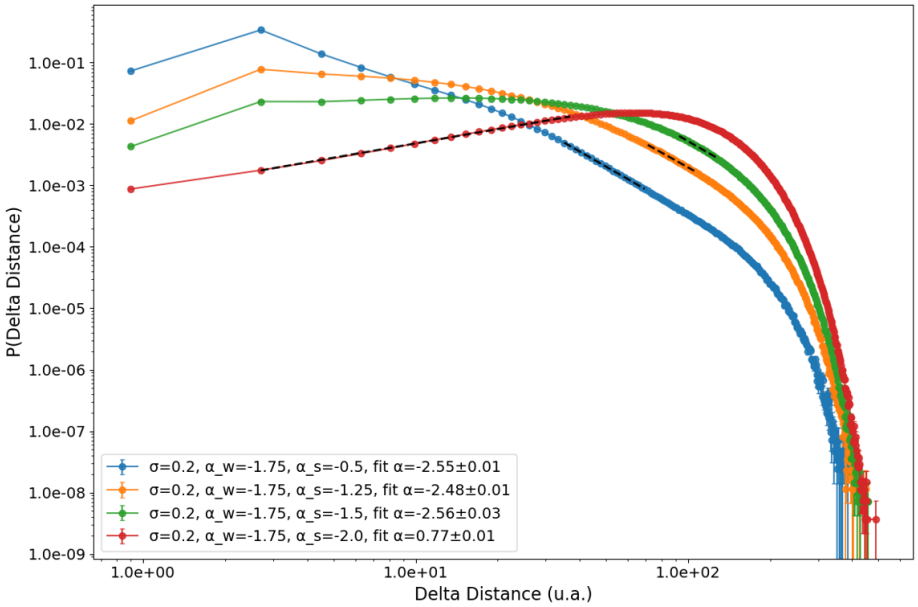


Figure S13: 15-minute displacement distribution for the third model, with $\sigma_\theta = 0.2$, $\alpha_W = 1.75$ and different values of α_S.